Wave Reflection and Transmission at Oblique Incidence: Parallel Polarization – Part 2

James T. Aberle, Ph.D.
Associate Professor Emeritus

School of ECEE

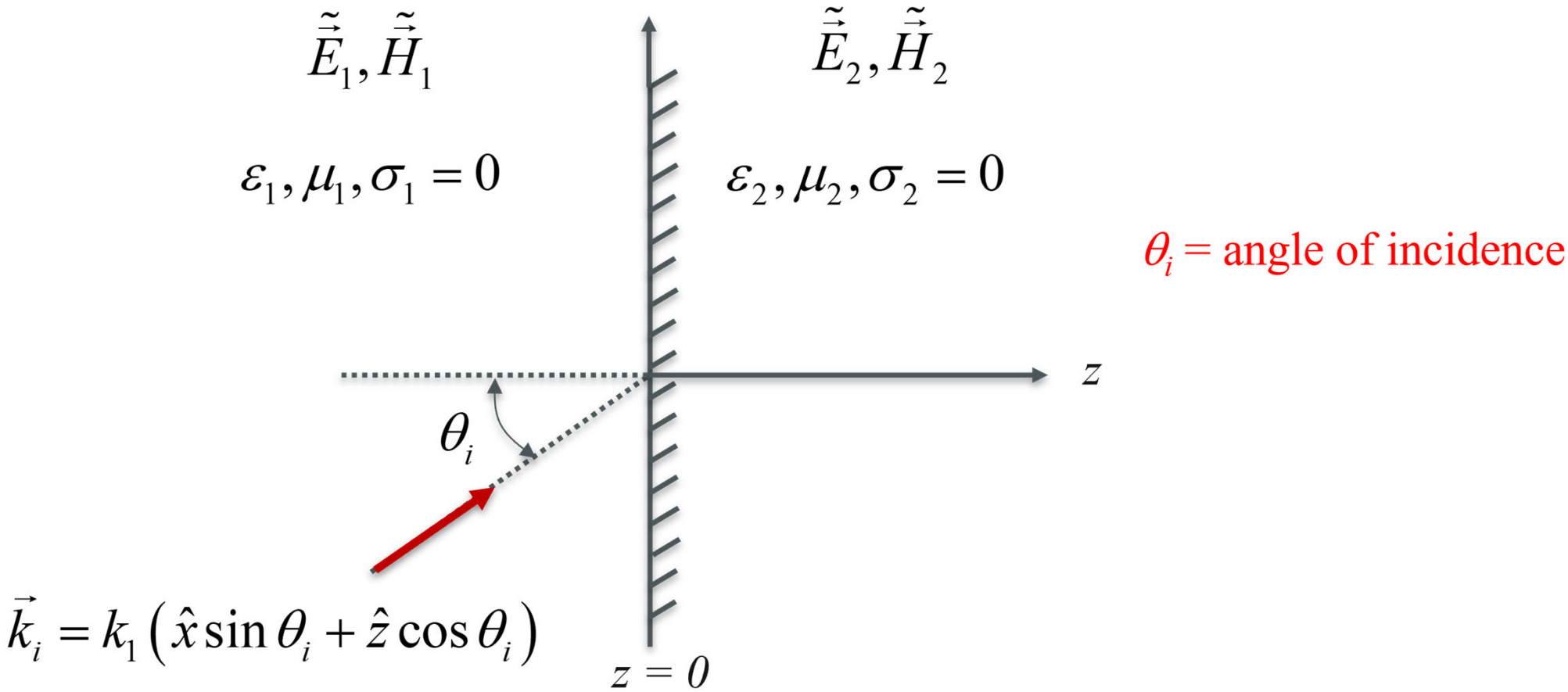
IRA A. FULTON SCHOOLS OF engineering
ARIZONA STATE UNIVERSITY

Lecture Objective

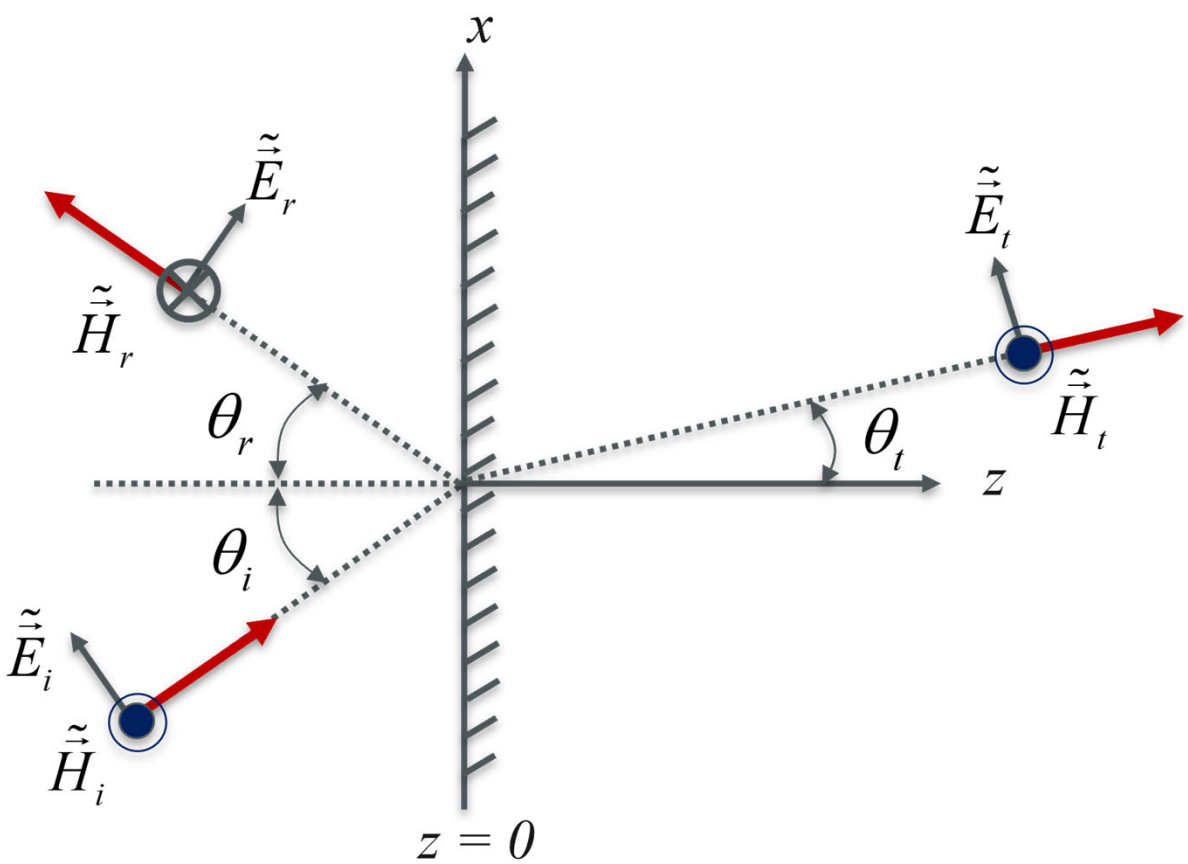
By the end of this lecture, you will be prepared to:

- Analyze plane wave reflection at the boundary between two dielectric half-spaces at oblique incidence for a wave with parallel polarization.

Oblique Incidence On a Plane Dielectric Boundary



Incident, Reflected and Transmitted Waves for Parallel Polarization



θ_i = angle of incidence

θ_r = angle of reflection

θ_t = angle of transmission

$$\vec{k}_i = k_1\left(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i\right)$$

$$\vec{k}_r = k_1\left(\hat{x}\sin\theta_i - \hat{z}\cos\theta_i\right)$$

$$\vec{k}_t = k_2\left(\hat{x}\sin\theta_t + \hat{z}\cos\theta_t\right)$$

Incident Wave for Parallel Polarization

- The magnetic field of the incident wave is given by

$$\tilde{\vec{H}}_i = \hat{y}\frac{E_0^i}{\eta_1}e^{-j\vec{k}_i\bullet\vec{R}} = \hat{y}\frac{E_0^i}{\eta_1}e^{-jk_1\left(x\sin\theta_i + z\cos\theta_i\right)}$$

- The electric field of the incident wave is given by

$$\begin{aligned}\tilde{\vec{E}}_i &= -\eta_1(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i)\times\tilde{\vec{H}}_i \\ &= (\hat{x}\cos\theta_i - \hat{z}\sin\theta_i)E_0^i e^{-jk_1(x\sin\theta_i + z\cos\theta_i)}\end{aligned}$$

Reflected Wave for Parallel Polarization

- Postulate that the magnetic field of the reflected wave is given by

$$\tilde{\vec{H}}_r = -\hat{y}\frac{E_0^r}{\eta_1}e^{-j\vec{k}_r\bullet\vec{R}} = -\hat{y}\frac{E_0^r}{\eta_1}e^{-jk_1\left(x\sin\theta_r - z\cos\theta_r\right)}$$

- The electric field of the reflected wave is given by

$$\begin{aligned}\tilde{\vec{E}}_r &= -\eta_1(\hat{x}\sin\theta_i - \hat{z}\cos\theta_i)\times\tilde{\vec{H}}_r \\ &= (\hat{x}\cos\theta_i + \hat{z}\sin\theta_i)E_0^r e^{-jk_1(x\sin\theta_i - z\cos\theta_i)}\end{aligned}$$

Transmitted Wave for Parallel Polarization

- Postulate that the magnetic field of the transmitted wave is given by

$$\tilde{\vec{H}}_t = \hat{y}\frac{E_0^t}{\eta_2}e^{-j\vec{k}_t\cdot\vec{R}} = \hat{y}\frac{E_0^t}{\eta_2}e^{-jk_2\left(x\sin\theta_t + z\cos\theta_t\right)}$$

- The electric field of the transmitted wave is given by

$$\begin{aligned}\tilde{\vec{E}}_t &= -\eta_2(\hat{x}\sin\theta_t + \hat{z}\cos\theta_t)\times\bar{H}_t \\ &= (\hat{x}\cos\theta_t \\ &- \hat{z}\sin\theta_t)E_0^t e^{-jk_2(x\sin\theta_t + z\cos\theta_t)}\end{aligned}$$

Reflected and Transmitted Waves for Oblique Incidence On a Plane Dielectric Boundary

- To uniquely specify the reflected and transmitted waves in terms of the incident wave, we need to evaluate:
  - E_0^r
  - E_0^t
- These unknowns are obtained by enforcing continuity of the tangential electric and magnetic fields across the boundary at $z = 0$ for all values of x.

Reflected and Transmitted Waves for Parallel Polarization

- Continuity of tangential electric field:

$$E_{1x}(x,0) = E_{2x}(x,0)$$
$$E_0^i \cos\theta_i \, e^{-jk_1 x \sin\theta_i} + E_0^r \cos\theta_i \, e^{-jk_1 x \sin\theta_i}$$
$$= E_0^t \cos\theta_t \, e^{-jk_2 x \sin\theta_t}$$

- Continuity of tangential magnetic field

$$H_{1y}(x,0) = H_{2y}(x,0)$$

$$\frac{E_0^i}{\eta_1} e^{-jk_1 x \sin\theta_i} - \frac{E_0^r}{\eta_1} e^{-jk_1 x \sin\theta_i} = \frac{E_0^t}{\eta_2} e^{-jk_2 x \sin\theta_t}$$

Reflection and Transmission Coefficients for Parallel Polarization

- Reflection coefficient:

$$\Gamma_{\parallel} = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$$

Fresnel reflection and transmission coefficients for parallel polarization.

- Transmission coefficient:

$$\tau_{\parallel} = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$$

Question

- Show that for $\theta_i = 0$ we get back the reflection and transmission coefficients for normal incidence.

Summary

- A plane wave obliquely incident on the planar boundary between two dielectric half-spaces is partially reflected back into the first medium and partially transmitted into the second medium.
- The reflection and transmission coefficients depend on the polarization of the incident wave quantify how much reflection and transmission occurs.

Ira A. Fulton Schools of Engineering

Arizona State University